

How to cross-compile device driver for rpi?

Step1: Download Linux kernel (source code) for raspberry pi from github.com in Ubuntu.

Link: <https://github.com/raspberrypi/linux>

Step2: Cross-compile the Linux kernel (source code) using cross-compiler in Ubuntu.

Step3: After cross-compilation, zImage (kernel image) gets generated in Ubuntu.

Step4: Send zImage from your Ubuntu to rpi.

```
scp zImage pi@ip_addr:/home/pi
```

step5: In rpi, mount bootfs partition in /tmp dir.

```
$ sudo mount /dev/mmcblk0p1 /tmp
```

```
$ cd /tmp
```

Step6: Copy zImage (which is received from Ubuntu) to kernel7.img.

```
$ cd /tmp
```

```
$ cp /home/pi/zImage ./kernel7.img
```

Imp Note: Before replacing kernel in rpi, take the backup of kernel7.img